Homework Answers

1. Bernie vs. a 150 Mbps transmission line

Using $1\,\text{GB} = 10^9$ bytes, Bernie carries

$$3 \times 7\,\text{GB} \times 8 = 168\,\text{Gb}.$$

At $18\,\text{km/h} = 0.005\,\text{km/s}$, the travel time for distance d is $d/0.005$ s. Bernie is faster when

$$\frac{168 \times 10^9}{d/0.005} > 150 \times 10^6 \quad \Rightarrow \quad d < 5.6\,\text{km}.$$

Thus, Bernie has the higher data rate for $0 < d < 5.6\,\text{km}$ (one-way distance).

- (i) If Bernie's speed doubles: $0 < d < 11.2\,\text{km}$.
- (ii) If each tape's capacity doubles: $0 < d < 11.2\,\text{km}$.
- (iii) If the line rate doubles to 300 Mbps: $0 < d < 2.8\,\text{km}$.

2. Layered protocols

Two reasons:

1. They divide a complex networking task into manageable modules.
2. A layer can be changed or replaced without redesigning the other layers.

Disadvantage: layering can add overhead and reduce performance.

3. Retransmissions

Each attempt uses one transmission; with probability p, another attempt is needed. Assuming independent errors and $p < 1$,

$$E[N] = 1 + pE[N] \quad \Rightarrow \quad E[N] = \frac{1}{1-p}.$$

The mean number of transmissions is $\frac{1}{1-p}$.

4. TCP vs. UDP

TCP is connection-oriented and provides reliable, ordered delivery. **UDP** is connectionless and provides no delivery, ordering, or retransmission guarantees.

5. Length of one bit in 802.3

The propagation speed is

$$v = \frac{2}{3}(3 \times 10^8) = 2 \times 10^8 \,\text{m/s}.$$

At 10 Mbps, one bit lasts $1/(10 \times 10^6)$ s, so

$$L = v \cdot \frac{1}{10 \times 10^6} = 20 \,\text{m}.$$

One bit was 20 meters long.

6. International standards

- Advantage: they ensure interoperability between equipment from different countries and vendors.
- Disadvantage: reaching agreement can be slow, and the resulting standards may be difficult to update.

7. Common OSI and TCP/IP layers

Both models have **application**, **transport**, and **network** layers (called **Internet** in TCP/IP). In the four-layer TCP/IP model, the **network access** layer covers OSI's data link and physical layers.